

Design and Implementation of a Web System for Searching and Managing Courses - Course Finder

First Report

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Abstract

Students often must search through numerous of courses, multiple instructors, and changing schedules across semesters, but information is often not centralized or in a palatable format, or in partially synchronized systems, making discovery and planning difficult. This project proposes a web-based system that consolidates course data and formalizes instructor-course relationships, separating authentication/authorization services from central domain-based course services, with non-functional objectives to integrate and centralize data, efficient search, and course-instructor associations. Workshops 1 and 2 established the central problem using business model canvas, user stories, CRC cards, and overall system design. Guided by Scrum as a development management methodology, preparing for sprint planning and traceability, this first report summarizes preliminary results and describes the project roadmap.

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1 Introduction

1.1 Context and Motivation

Students often have no way to organize their semester schedules. At many institutions, including our own, course information is often poorly organized. This fragmentation makes it difficult for students to discover relevant offerings of interest, as well as the professors who teach them and their schedules. The result is avoidable problems during planning and registration periods, duplicate registrations, and delays in decision-making.

To address these issues, we propose a web system for searching and managing academic courses that consolidates essential data (course codes, titles, descriptions) and formalizes the relationships between professors and courses. The system targets three primary stakeholder groups: (i) *students*, who need fast, accurate search and discovery; (ii) *professors*, who is the associated with the courses; and (iii) *administrators users*, which is responsible for loading the data and managing the application. Our current work established the problem framing and initial design assets for the development phase.

From an engineering perspective, the project is intentionally designed to practice delivering end-to-end software within a semiannual schedule by implementing an agile approach.

1.2 Objectives

- Design and specify a reliable web-based system that allows key stakeholders to search, manage and maintain accurate information about courses and instructors.
- Define a clear and feasible roadmap for the implementation of the project that aligns with the identified problem context, stakeholder needs, and planning.
- Synthesize the context, stakeholders, and scope of the problem into a consistent baseline that unifies terminology and guides subsequent design and development activities.

1.3 Scope

This first report defines the requirements and design boundaries of the project at its current stage. It focuses on consolidating the conceptual and architectural foundations, including problem definition, stakeholder identification, user stories, CRC cards, high-level architecture, deployment views, and early user-interface sketches. The report also introduces preliminary elements for the data model and API contracts that will guide the implementation in the next phase. This scope explicitly includes documenting the problem context, justifying the proposed solution, and establishing an initial roadmap under agile methodology using Taiga as the project management tool.

2 Related Work and Background

Students need digital systems to search and manage academic information such as course offerings, instructor assignments, and scheduling. However, on many occasions they find that in fragmented data or centralized in legacy applications that lack interoperability and transparency for stakeholders. In this context, academic management systems and course catalog platforms such as Moodle, Blackboard, and Google Classroom provide partial solutions, focusing primarily on learning management rather than the structural organization of academic offerings.

From a technical point of view, modern web-based architectures rely on RESTful APIs, microservice decomposition, and standard authentication mechanisms such as JWT (JSON Web Token) to ensure secure and maintainable information exchange. These principles underpin the design of this project: a lightweight, modular system divided into a authentication service and a business service. This dual-back-end structure supports scalability, separation of concerns, and parallel development by different team members. Furthermore, adopting a continuous integration and deployment pipeline through tools like Docker and GitHub Actions allows the project to align with current DevOps practices that promote automation, repeatability, and early defect detection.

In terms of methodology, the use of Scrum supported support on the Taiga tool reflects an agile approach, iterative development of software engineering. Agile methods are widely used in both industry and academia to encourage adaptability, transparency, and stakeholder feedback. Previous studies and projects in software engineering have shown that integrating agile practices allows for better quality deliveries.

In summary, this work draws on principles from web-based academic management systems, requirement analysis and software design to propose a simple, fast, and verifiable solution tailored to the needs of the stakeholders.

3 Methodology

3.1 Process and Team Practices

The project will follows an agile software development approach based on the Scrum framework, adapted to the context of problem. The team uses **Taiga** as the main project management tool to maintain the product backlog, define sprint goals, and track progress through user stories and tasks

Roles are distributed collaboratively among the three members of the team. Carlos Andres Pescador Castro serve as the Product Owner and is the responsible for prioritizing the backlog and maintaining alignment with the defined requirements. Cristian Santiago Ríos Vásquez act as the Scrum Master, facilitating coordination among team members,

ensuring adherence to Scrum events, and managing sprint reviews and retrospectives. Brayan Alejandro Riveros Rodríguez as the Development Lead, focusing on technical implementation, version control integration, ci/cd pipeline and testing coordination. All members contribute equally to documentation, gathering of requirements and continuous improvement activities.

Scrum ceremonies are conducted at a scaled frequency appropriate brief stand-up meetings are held twice per week (asynchronous when needed), sprint planning and retrospectives occur at the beginning and end of each workshop, and sprint reviews coincide with workshop submissions. Version control and collaboration are managed through a shared Git repository, following a feature-branch workflow.

3.2 System Analysis and Design

Following the completion of Workshops 1 and 2, the system analysis and design phase guide the transition from requirements specification to technical implementation. The main goal will be to ensure that the proposed architecture, domain entities, and component responsibilities are consistent with the user stories, acceptance criteria, and stakeholder needs identified during the analysis stage.

The analysis focus on consolidating the functional requirements derived from the user stories created in Workshop 1. Each user story will be mapped to a corresponding set of system components, defining clear inputs, processes, and outputs. CRC (Class-Responsibility-Collaborator) cards previously designed will serve as the foundation for assigning responsibilities among classes, while maintaining object-oriented design principles. This mapping will guarantee that each functional requirement has a direct representation within the software model.

The system will adopt a modular architecture divided into two main backend services: an authentication and authorization service, and a core domain service for managing courses, professors, and their relationships. Both services will communicate through RESTful APIs, enabling independent deployment and scalability. Each backend will interact with its respective database—MySQL for authentication and PostgreSQL or MongoDB for the core domain—ensuring persistence and referential integrity across entities. The frontend web interface, developed with modern JavaScript framework, will consume these APIs to provide a cohesive user experience.

To formalize the design, UML diagrams are used to illustrate the system structure and interactions. A class diagram will describe the main domain entities (Professor, Course, and Link), their attributes, and associations. A deployment diagram will detail how components will be distributed across environments, showing the web client, API services, and database instances.

Non-functional requirements will also guide the design. The system will ensure se-

curity through JWT-based authentication and role-based access control, maintainability through modular coding practices, and performance through optimized database queries and API endpoints. Furthermore, logging and basic monitoring mechanisms will be defined to support future testing and evaluation.

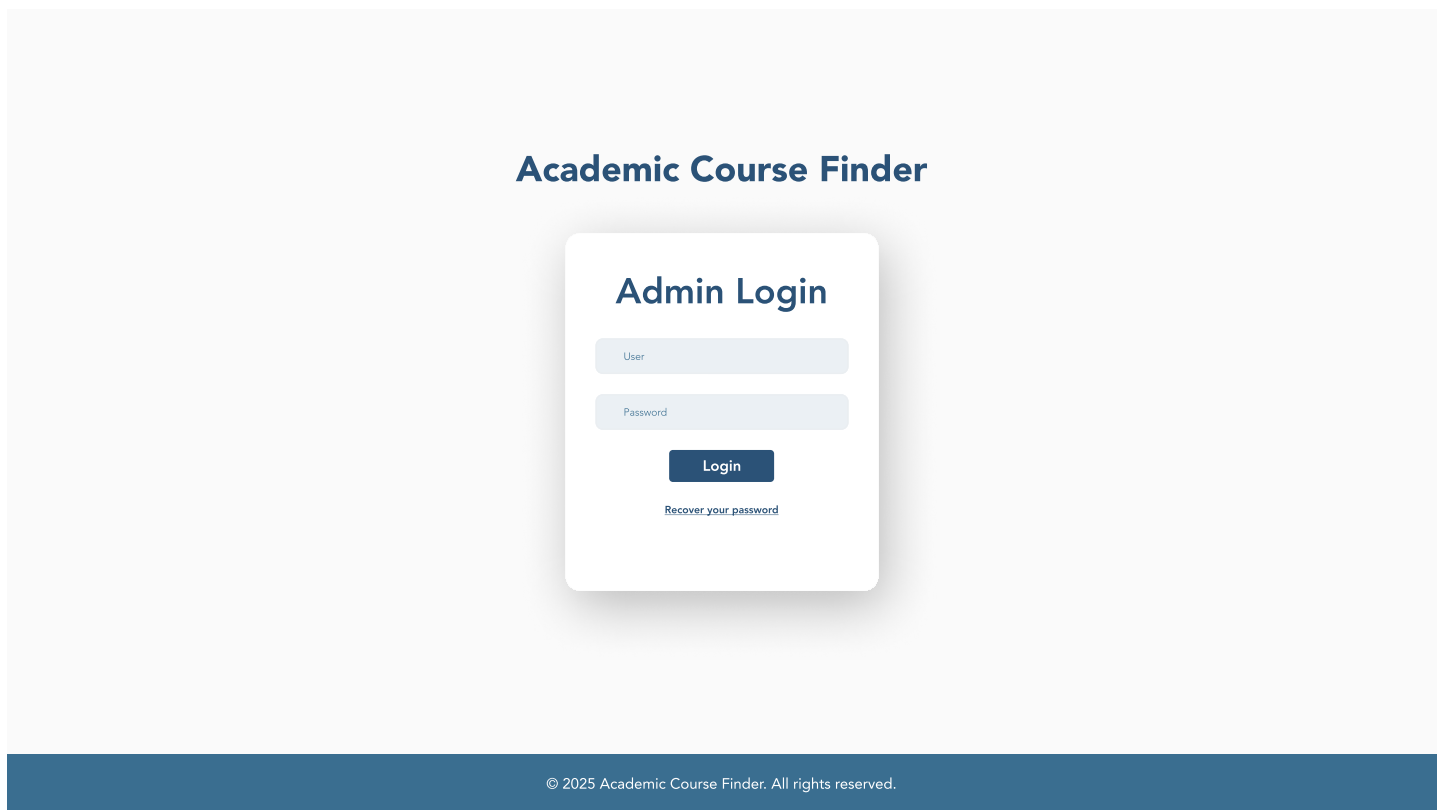
This analysis and design process will establish a clear technical blueprint for the implementation stage to be developed in Workshop 3. By grounding each architectural decision in the previously validated requirements, the team will ensure traceability from stakeholder needs to system components and a solid foundation for the subsequent development and testing phases.

4 Implementation Progress (Workshops Summary)

4.1 Web UI

Early screens for search, results, admin login and flows are drafted.

Figure 1: Web UI Mockup (v1.0)



The image is a web UI mockup for an 'Academic Course Finder' application. It features a central white login card on a light gray background. The card is titled 'Admin Login' in a bold, dark blue font. Below the title are two input fields: 'User' and 'Password', both with light gray borders and placeholder text. A dark blue 'Login' button is positioned below the password field. Underneath the button is a link that reads 'Recover your password' in a small, dark blue font. The entire mockup is set against a light gray background, and a dark blue footer bar at the bottom contains the copyright text: '© 2025 Academic Course Finder. All rights reserved.'

Academic Course Finder

Admin Login

User

Password

Login

[Recover your password](#)

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Figure 2: Web UI Mockup (v1.0)

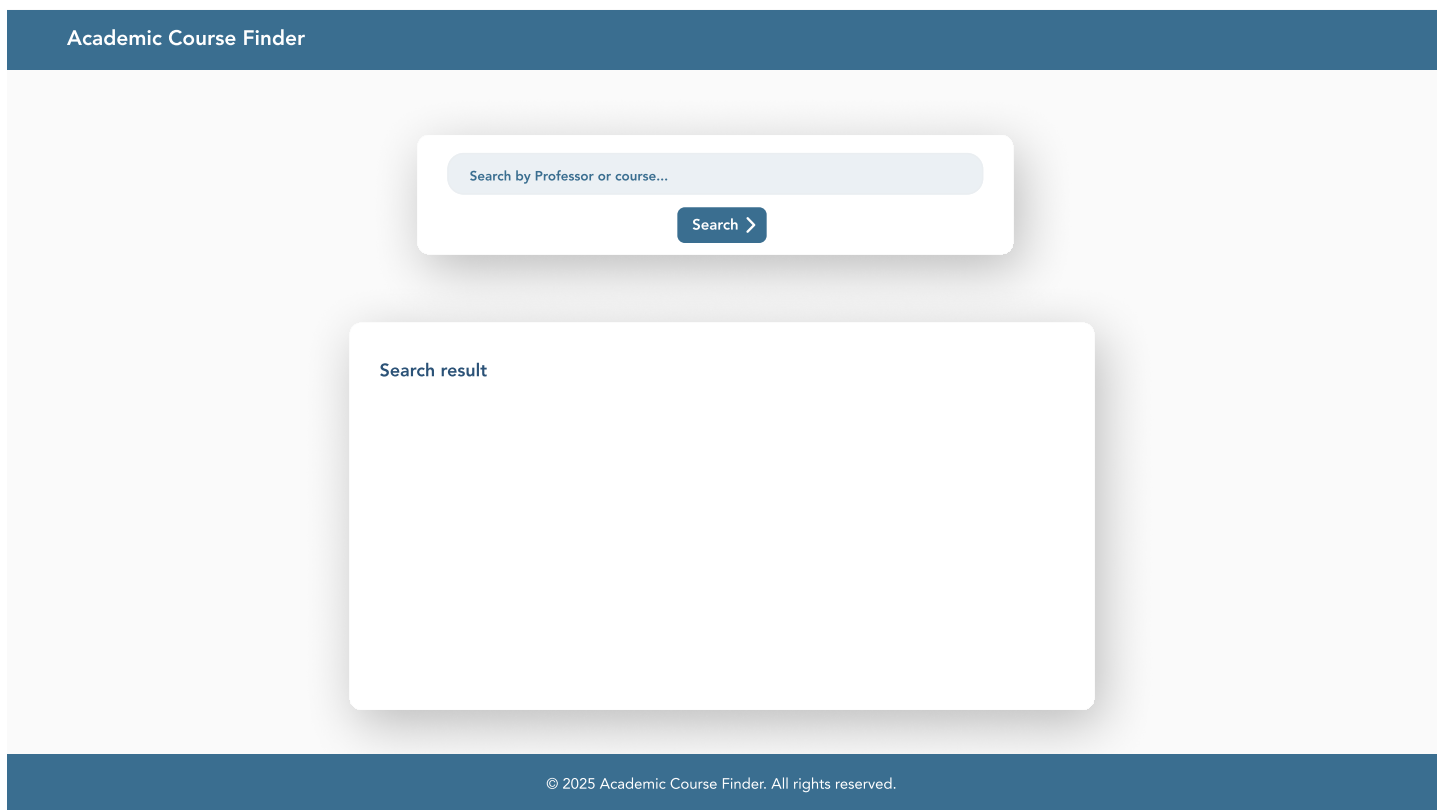


Figure 3: Web UI Mockup (v1.0)

Academic Course Finder

Search by Professor or course...

Search >

Professors teaching Calculus

Name: -----

Code: -----

Email: -----

Name: -----

Code: -----

Email: -----

Name: -----

Code: -----

Email: -----

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Figure 4: Web UI Mockup (v1.0)

Academic Course Finder

Search by Professor or course...

Search >

Courses taught by: Dr. Jhon Smith

Name: -----
Group: -----
Schedule: -----

Name: -----
Group: -----
Schedule: -----

Name: -----
Group: -----
Schedule: -----

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Figure 5: Web UI Mockup (v1.0)

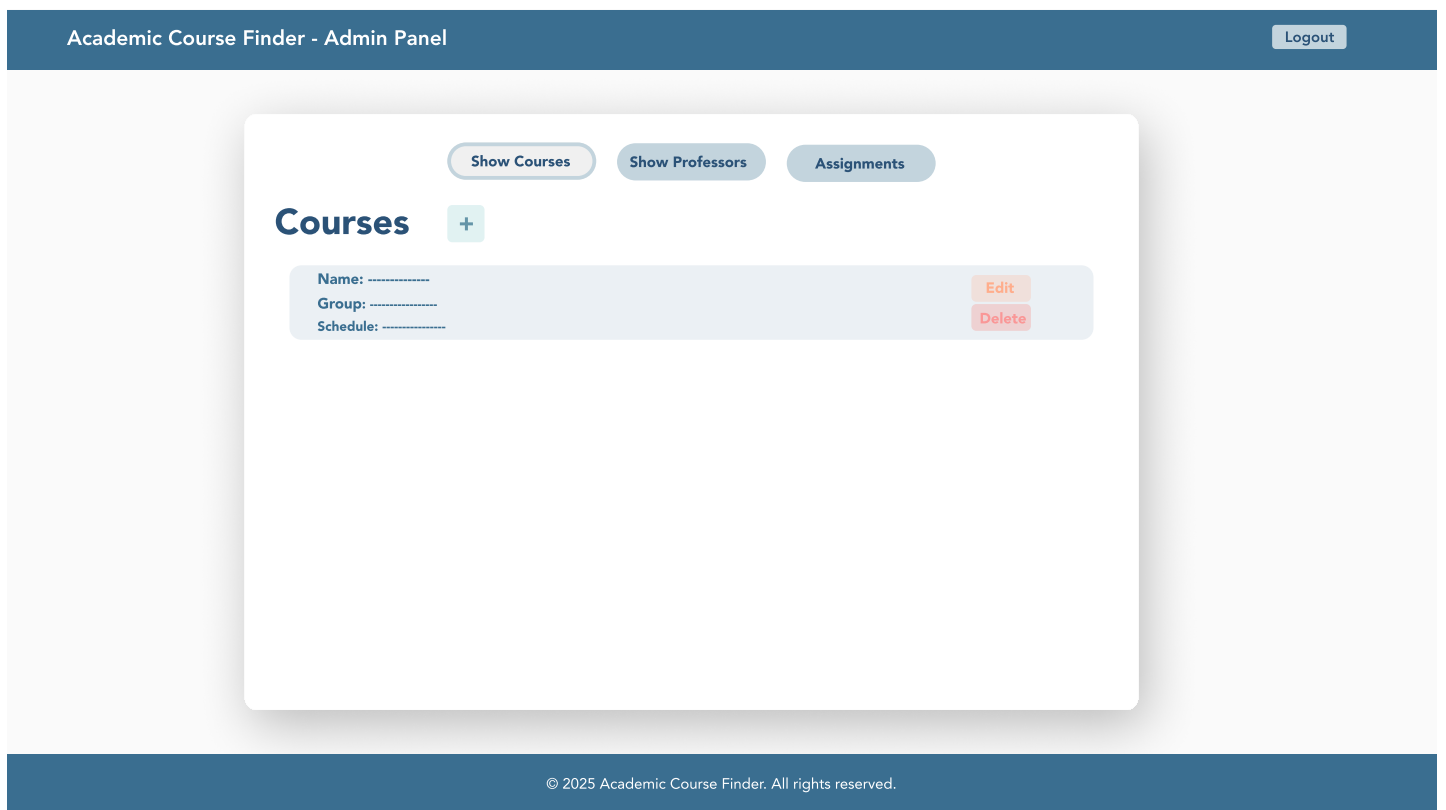
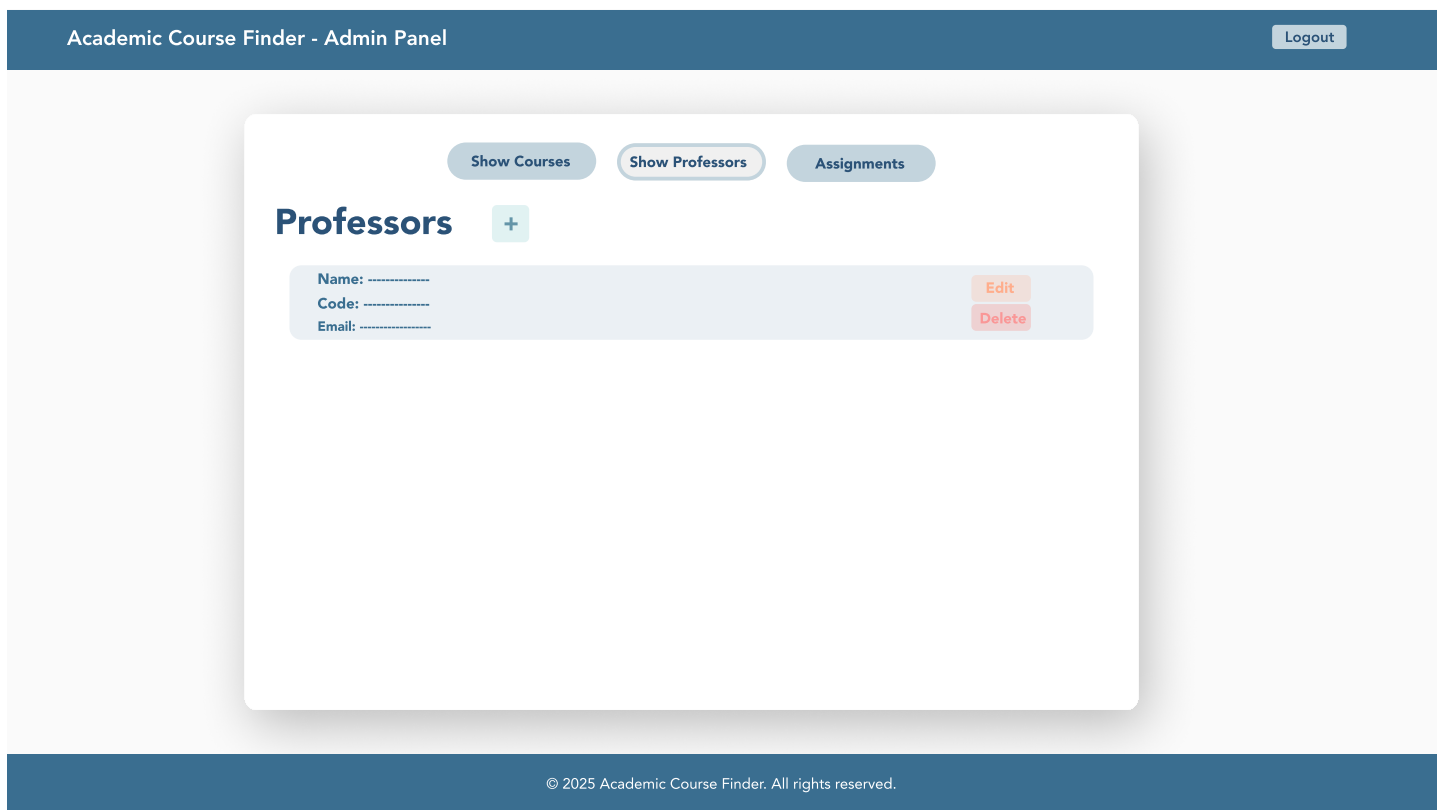


Figure 6: Web UI Mockup (v1.0)



4.2 Architecture Overview

High-level architecture

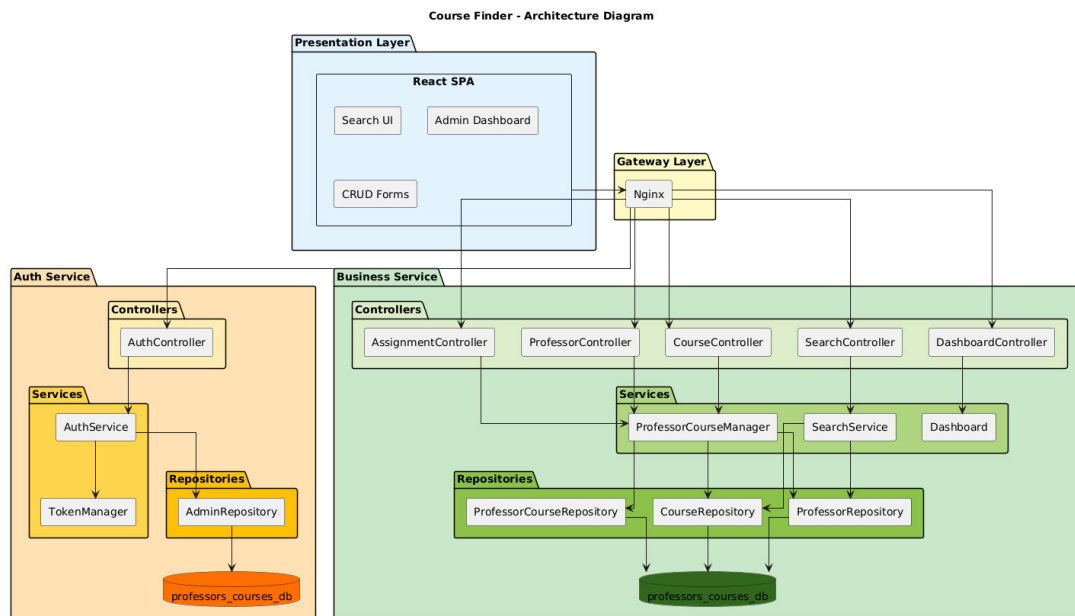


Figure 7: Architecture Diagram (v1.0).

4.3 Deployment View

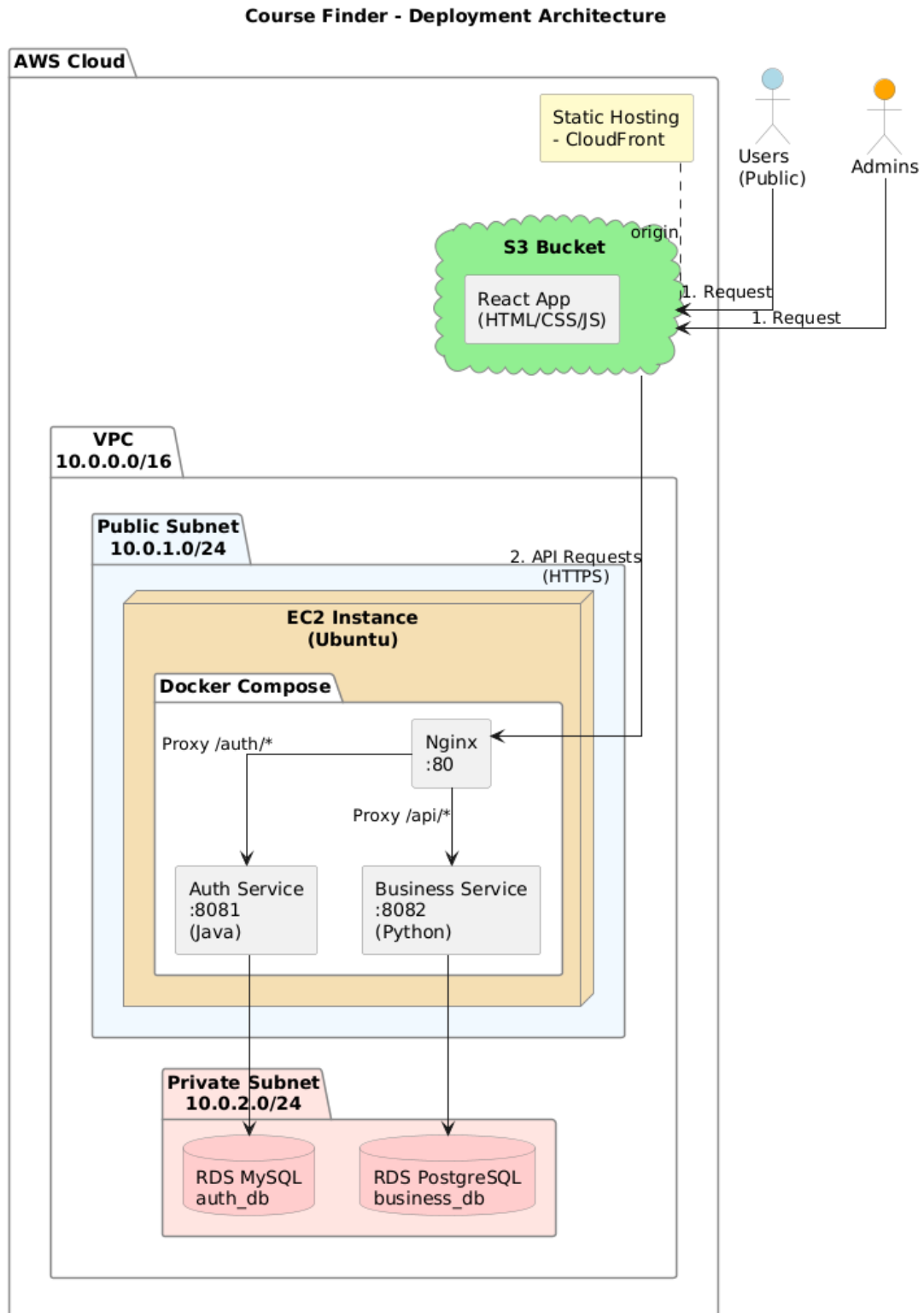


Figure 8: Deployment diagram (v1.0).

4.4 Data Model

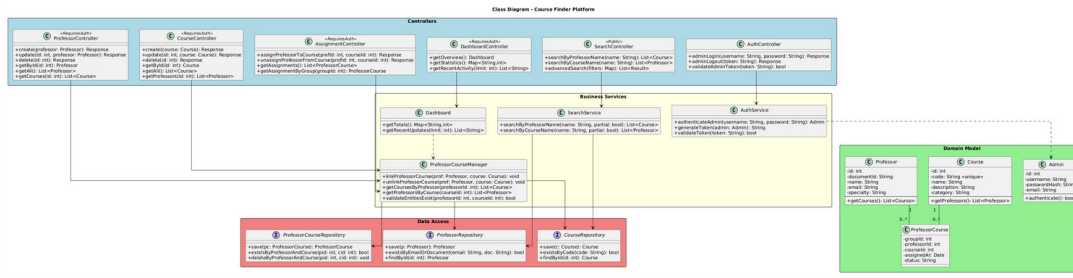


Figure 9: Class diagram (v1.0).

5 Results and Discussion

The outcomes of Workshops 1 and 2 have established a solid analytical foundation for the project. The main deliverables include the problem definition, user stories with acceptance criteria, CRC cards, architectural and deployment diagrams, and early Web UI mockups. These artifacts provide a coherent view of the system's objectives and structure, enabling the team to validate the feasibility of the proposed solution before implementation.

6 Conclusions

This first report consolidates the analytical and design foundations of the project, demonstrating coherent progress from requirement elicitation to system architecture definition. The established artifacts and planning ensure a clear path toward implementation, testing, and integration in the following workshop stages.

Glossary

API Application Programming Interface.

CI/CD Continuous Integration / Continuous Deployment.

JWT JSON Web Token.

CRC Class - Responsability - Collaboration Cards